LeepTalent Level 3 End-point Assessment

Data Technician

SAMPLE

Assessment Method:

Scenario Demonstrations with Questioning

90 minutes (for Scenario 1 and 2)

Scenario Demonstration 1 – Data Gathering and Validation

45 minutes

You are employed as a data technician by an economic research organisation. The organisation is reviewing long-run UK employment by industry and has asked you to prepare a single, clean data set that combines the men's and women's employment series, ready for analysis.

You have been provided with the following data sources: **Employment by industry - men.csv**, **Employment by industry - women.xlsx**, and the **Quarter-to-Date** reference table in **Annex A**. Using a suitable data analysis tool of your choice, carry out the following:

Task 1 — Import the data sources

- Import both employment files into your chosen tool.
- Review the column names, formats and structure of each to confirm they are compatible — note that the source files carry several header rows and grouping information above the data.

Task 2 — Combine the data sources

- Combine the men's and women's series into a single data set.
- Ensure column names are consistent, and add a **Gender** column to identify which series each record came from.

Task 3 — Validate and clean the data

- Check the combined data set for errors (negative or impossible values, misspellings), inconsistencies in naming, and duplicate rows.
- Remove any unnecessary rows or columns, and note that early periods use a placeholder (**..**) where no sector breakdown is available.
- Set the correct data type for each field: dates as dates, employment figures as numbers, industry names as text.

Task 4 — Standardise the date field

- Convert the quarter field into a full date using the **last day of the quarter**, as set out in **Annex A** (for example, **Jan-Mar 1995** → **31/03/1995**).

Save the prepared data set with a clear name (for example, **Employment_Cleaned.xlsx**).

Scenario Demonstration 2 – Data Analysis

45 minutes

Using the cleaned employment data set from Scenario Demonstration 1, carry out the following analysis and present your findings in a format suitable for senior management. Where a time range is needed, use **1997–2023**, as the sector breakdown begins in 1997.

Task 1 — Public and private sector trends (1997–2023)

- Analyse employment trends in the **public** and **private** sectors over time.
- Create tables and charts showing the changes, and summarise with key statistics (average, maximum, minimum).

Task 2 — Industry focus: Mining, energy and water supply

- Extract the **Mining, energy and water supply** series and visualise its trend from 1997 to 2023.
- Compare male and female employment in this sector.

Task 3 — Gender-based comparison

- Compare employment trends for men and women across the full range, both overall and for selected sectors (e.g. manufacturing, education, health).
- Use side-by-side bar charts and line graphs.

Task 4 — Trends and patterns

- Identify any seasonal patterns, sudden rises or falls (e.g. around economic events), and any convergence or divergence between male and female employment over time.

Task 5 — Correlation analysis

- Explore relationships between sectors — for example, whether a rise in private sector employment relates to a change in the public sector — using correlation and scatter plots.

Task 6 — Forecast to 2028

- Use the existing trends to forecast employment for the next five years, for overall employment and for one or more selected sectors, using a suitable method (e.g. linear regression or a forecast tool).

Annex A – Quarter-to-Date Reference

Use this table to convert each quarter label to the last day of that quarter in Scenario 1, Task 4. Append the four-digit year from the quarter label.

Quarter label	Last day of quarter	Example
Jan-Mar	31/03/YYYY	Jan-Mar 1995 → 31/03/1995
Apr-Jun	30/06/YYYY	Apr-Jun 2002 → 30/06/2002
Jul-Sep	30/09/YYYY	Jul-Sep 2010 → 30/09/2010
Oct-Dec	31/12/YYYY	Oct-Dec 2018 → 31/12/2018